

Pre-Calibration Set-Up and Troubleshooting

Tips

In this document, you find direct links to posts that will provide answers to the most frequently asked questions, and tips for troubleshooting various issues that arise when calibrating LG OLED displays.

Please take the time to read through the FAQ's and the provided links.

OLED displays are considered the pinnacle of consumer current TV technology. Thanks to the internal LUT capabilities of LG OLED, when calibrated they can produce a true “reference” image when displaying SDR content. But there are some quirks when calibrating a consumer (WRGB) OLED display, and not all OLED panels are created equal.

I have already included a lot of information in the 3D LUT Calibration Guide, and although the guide was created as a more step-by-step walkthrough and contains links to such things as Pre-Calibration steps, it seems logical to provide a simplified sheet with answers to the most FAQ's.

Please take the time to read through the FAQ's and the provided links.

Q: What is ColourSpace?

A: ColourSpace is a new calibration software from Light Illusion. Although it is currently functionally similar to its predecessor (LightSpace), it is a totally new software that has been built from the ground up to allow for future upgrade possibilities. It also has a much easier user interface and is much more flexible. LightSpace was already flexible, but could be somewhat cumbersome to use at times. There are no such issues with ColourSpace, and your calibration workflow is only limited by your imagination.

[Light Illusion – ColourSpace main page](#)

Q: So, can I use AutoCAL with ColourSpace?

A: No. The AutoCAL function used by alternate calibration solutions initially uses a form of automated manual adjustment of specific points of the greyscale which it then calculates a 1D LUT from. This is followed by the

3D LUT which can use either a pre-determined fixed number of points, or an “intelligent” randomly generated set of points (time or point based set by the user).

The problem is that all LG OLED’s suffer issues when using the custom 1D LUT slot, so as soon as you click that AutoCAL button you have already made the potential outcome worse than it would be if you manually adjust the greyscale from the user menu. In addition to this, the 3D LUT generation provided by alternate solutions does not perform well with the non-additive characteristics (WRGB) of LG OLED. This causes image artefact issues such as posterization and banding, especially in low light conditions (an area where OLED should excel). Unfortunately, this happens even with the (quick) Lightning LUT method which simply measures 26 steps of White and 25 of Red, Green and Blue and calculates a full 33x33x33 Calibration LUT (35,937 correction points). This is not really a viable way to calculate a 3D LUT as it needs to contain volumetric data of as many possible colours at as many levels of intensity as possible to be considered accurate, especially with a non-additive (WRGB) display. The minimum recommended measurement set to provide full accuracy is a 17x17x17 sequence. That is 4913 measured points (far more than the 101 points measured using Lightning LUT).

Q: The LG OLED has 1D and 3D LUT slots for each picture mode, why do you skip the 1D LUT.

A: As stated above, the LG has some issues with the 1D LUT processing. These vary by picture mode (SDR/HDR/Dolby Vision) but each issue becomes noticeable as soon as you reset the 1D LUT slot.

When you reset the 1D LUT, you are bypassing specific processing functions of the TV. Values such as Brightness, Contrast and Colour **must** remain at their default values for the points contained within the 1D LUT to align correctly. The 1D LUT will attempt to correct white balance and gamma to your target from the native colour temperature and factory stored (approximate) 2.2 gamma of the panel. With the 1D correction LUT loaded, basic measurements will make it seem as if everything is perfect, but on further evaluation with test patterns and more advanced measurements and graphs you will discover the issues. Raised/crushed blacks (varies per picture mode). Banding, specifically in the lower regions of a grey scale ramp. Clipping of colour.

This is one of the reasons you sometimes see users posting their “outstanding” results in an AVS Forum post one day, only to return a short while after complaining about issues and seeking assistance. Some of this can be exacerbated or somewhat masked by panel variance, but it is definitely part of the equation as to why so many users have issues.

Q: So, what is the solution to this issue?

A: Skip the 1D LUT entirely!!!!

Q: But if you skip one part of the calibration, how can you get good results?

A: The answer is not only very simple, but actually very logical when you think about it.

A large volumetric measurement profile will not only measure a series of colours at various saturations and luminance values, but it will also measure multiple steps within the grey scale. This allows the calculation of the 3D LUT to correct not only for colour accuracy, but for white balance and gamma too.

Q: If the 3D LUT corrects white balance, why pre-calibrate white in the Service Menu?

A: Again, the answer is both simple and logical.

It is entirely possible to leave the Service Menu White Balance settings as they are from the factory. It is also entirely possible to set all of the values to 192.192.192 in the Service Menu which will be the native colour temperature of the panel. You can then measure your native panel colour temperature and native panel gamma response and allow the 3D LUT correct everything. And it will work very well.

But doing this will make it more difficult to set your peak output level. You will have to set your initial peak output significantly higher than the target you are aiming for because as white balance is corrected, peak luminance will fall considerably.

You will also likely need to measure a much larger profile of your display, or at least add more grey scale points. This can be done either by manually adding them to a patch sequence, or augmenting a separate large grey scale measurement profile. Basically, it ends up over complicating a process for no good reason to end up with the same end results. Plus, the over complication adds a significant amount of time to the process.

The other benefits of pre-calibrating white in the Service Menu are that the adjustments will not only correct the white balance of the entire greyscale very well, they will also apply to all picture modes to have a solid base to work from, including HDR and Dolby Vision.

Q: Got it, pre-calibrating white has benefits. How do I do it?

A: Entering the Service Menu is a risk and should only be done after fully reading and understanding the procedure. If the instructions are followed correctly there is absolutely no risk at all, but unfortunately there is always that one person that think they know better and do not follow the instructions as provided. This then leads to constant support requests of things like “I followed the instructions and now {insert issue here}”. Well, I’m sorry to say this but if you end up with issues you did **not** follow the instructions.

Understand the procedure and the risks before proceeding. I accept no responsibility if you make a mistake!

One of the biggest tips I can give before you make changes is that you record the factory values of each colour temperature. You can write these down or simply just take a picture of each setting with a smartphone camera. You may never need them again, but it is good practice to retain them should the need ever arise to revert to factory calibration.

NEVER PRESS RESET WHILE IN THE SERVICE MENU WHITE BALANCE MENU

To Enter the Service Menu, you will need a remote that supports sending the IN-START and EZ-ADJUST commands. The LG Service Remote (Model code: MKJ39170828) and most Harmony Smart Remotes are suitable choices.

When entering the Service Menu, enter 0413 as the password.

Here are some links to posts by Ted Aspiotis on how to calibrate Service Menu White Balance

- [LG White Balance Pre-Calibration using Service Menu](#)
- [Peak White Pre-Calibration Tip - Before Starting the 3D LUT profiling](#)
- [Explanation as to why Service Menu White Balance calibration is performed](#)

!!!!!! NEVER EVER PRESS THE IN-STOP BUTTON – YOU HAVE BEEN WARNED !!!!!

Q: Why do I need to disable certain things for calibration?

A: Post processing of images affect the way images are displayed. When displaying calibration patches, we need the incoming signal to remain as pure and as unaltered as possible. Excessive Sharpness will add ringing artefacts, Motion Interpolation will add additional frames that aren't there. This will adversely affect the accuracy of any measurement taken. They must be disabled prior to calibration and "purists" will most likely leave them disabled forever. They can, however, be enabled after calibration if you so choose.

Things that must be disabled are as follows.

Global Picture settings:

- | | | |
|-------------------------|------|--|
| • Aspect Ratio | 16:9 | <i>Set Just Scan – On</i> |
| • Energy Saving | Off | |
| • Eye Comfort Mode | Off | <i>Under Additional Settings</i> |
| • HDMI Deep Colour | On | <i>Under Additional Settings</i> |
| • Instant Game Response | Off | <i>Under Additional Settings</i> |
| • Screen Shift | Off | <i>Under Additional Settings -> OLED Panel Settings</i> |
| • Logo Luminance | Off | <i>Under Additional Settings -> OLED Panel Settings</i> |

Picture Mode Settings:

- | | | |
|------------------------|-----|--|
| • Sharpness | 0 | <i>Default value of 10 actually adds some sharpness processing</i> |
| • Dynamic Contrast | Off | <i>Under Expert Controls</i> |
| • Super Resolution | Off | <i>Under Expert Controls</i> |
| • Peak Brightness | Off | <i>Under Expert Controls</i> |
| • Noise Reduction | Off | <i>Under Picture Options</i> |
| • MPEG Noise Reduction | Off | <i>Under Picture Options</i> |
| • Smooth Gradation | Off | <i>Under Picture Options</i> |
| • Black Level | Low | <i>Under Picture Options</i> |
| • Real Cinema | On | <i>Under Picture Options</i> |
| • Motion Eye Care | Off | <i>Under Picture Options</i> |
| • TruMotion | Off | <i>Under Picture Options</i> |

General Settings:

- | | | |
|-------------------------|-----|---|
| • Eco Mode | Off | <i>2018 and 2019 LG OLED's only</i> |
| • Auto Power Off | Off | <i>Under General/Additional Settings/Eco Mode – 2020 LG OLED's Only</i> |
| • Sleep Timer | Off | <i>Under General/Time – 2020 LG OLED's Only</i> |
| • Power On/Off Timer | Off | <i>Under General/Time – 2020 LG OLED's Only</i> |
| • AI Picture Pro | Off | <i>AI Service – 2020 LG OLEDs only</i> |
| • AI Brightness Control | Off | <i>AI Service – 2020 LG OLEDs only</i> |
| • Auto Genre Selection | Off | <i>AI Service – 2020 LG OLEDs only</i> |

As stated, many of these can be enabled after calibration at your own discretion, but they must be disabled beforehand.

Q: ASBL and GSR, what are they?

A: Both of these Service Menu settings are related to burn-in protection features of the TV.

ASBL (Named TPC in the Service Menu) will gradually dim the screen when a static image is displayed for a period of time. It does not just do this with bright images, but with dim ones too. So it can be beneficial to disable this, especially if you are using a consumer grade meter that takes longer to read dark patches.

GSR is a similar function that reduces the light output of static images (such as logos). This setting does not seem to affect calibration but can be disabled if so desired.

It must be emphasised that both ASBL and GSR are burn-in protection features and are disabled at your own risk. If unsure, they can be disabled prior to calibration and enabled again once complete.

To disable these features either permanently or temporarily, click the links below to see posts from Ted.

- [Disable ASBL -TPC](#)
- [Disable GSR](#)
- [Initial discovery of additional GSR Service Menu option](#)

Credit to AVS Forum user jk82

Q: Is that all I need to do before profiling the display?

A: In many cases then after white point and peak luminance has been accurately set, you should be set to fully profile your display for LUT Generation. Unfortunately, it is not always the case and this comes down to panel variance and how good the factory calibration of your specific panel is.

When we talk about the factory calibration of your panel, we are not implying that LG Electronics calibrates each panel individually. LG Electronics only calibrate one point of the Service Menu White Balance in an “automated” manner to three separate colour temperatures and then apply a generic colour transform matrix upon that for each picture mode.

The factory calibration of the panel we refer to is the one that LG Displays perform and store to the TCON unit of each individual panel. This factory calibration includes applying the 2.2 gamma response and affects how each panel comes out of black for example. This may mean one panel is perfectly fine with a Brightness value of 50, where another may need a value of 52, or even higher. The highest value I have personally witnessed is 56, but I have seen reports of users having to go as high as 60. Panel variance is real!!!

But it also affects the luminance range of the panel, which means on the brighter end it can cause clipping of colour with the default value of Contrast at 85.

What you can take from this is that if the default “bypass values” of Brightness, Contrast and Colour are left as is, then absolutely nothing you do with a 1D LUT can correct these issues. Another reason it is advised to skip either resetting or uploading a 1D LUT.

Fear not!!!

From one very quick and simple measurement profile you can discover if your particular panel suffers from these issues and can correct them very quickly and very easily by viewing the “RGB Separation Chart”.

Once again, here are some links to posts by Ted explaining the issue and tips on how to resolve them.

- [Explanation of the issue](#)
- [Explanation of RGB Separation](#)
- [RGB Separation Examples](#)
- [AVS Forum user – Issue solved](#)
- [Example from my own personal experience](#)

Q: I’ve fixed RGB Separation, what’s next?

A: Now you have ensured that your panel is optimally setup, profiling can begin. But this is where things can get a little bit tricky on an unstable display such as a WRGB OLED.

Due to the unstable nature of these panels, they can fluctuate quite dramatically, especially when displaying bright, high luminance images. This is the reason we recommend calibrating to the reference standard of 100 nits @100% White when performing your first full display profile. At 100 – 120 nits the panels remain relatively stable, although they can still drift a little.

Fortunately, due to the flexible nature of ColourSpace it is very easy to create your own profiling patch sequences, and this has been done for you by a very generous AVS Forum user and provided free of charge to all ColourSpace users.

The patch sequence you have been provided with is a 17x17x17 cube that has been sorted to retain an average patch power level to help combat display drift. This, in combination with the Pre-Roll (used to prepare the panel to that same patch power level) and the Drift Compensation patches inserted during profiling allows the LUT generation engine to calculate an accurate LUT. The verification patch sequence has been sorted in the same way.

Whilst these patch sequences are often referred to as “Special Patch Sets”, there is nothing really special about them other than that they have been optimally sorted for WOLED.

Q: But I don’t always view in a dark room and 100 nits is too dim!!

A: And that is totally fine. Nobody is telling you to restrict yourself to a single 100 nit picture mode. You can calibrate a day mode at 250 nits, or an early evening mode at 180 nits if you so choose. You can max out the OLED Light to 100, enable Peak Brightness High and calibrate that too. Just don’t expect it to be as accurate due to WOLED instability. We suggest, and highly recommend 100 nits to begin with so that users familiarise themselves with the software, the procedure and the limitations of WOLED in general.

Concentrate on your critical viewing mode first and foremost. When that is complete then calibrate your “day mode” at whatever target you like. Total accuracy is not required in this scenario due to ambient light contamination and other factors, so you can also use a smaller (faster) patch sequence too.

Q: Why can't I create a 3D LUT for HDR in ColourSpace?

A: You can. It just won't be any good!

LG OLED actually has available the same 33x33x33 LUT container it uses in SDR in HDR too. It is, in the most basic terms, unusable!

Again, this mainly comes down to panel instability but other factors come into play such as the severe loss of colour volume at HDR luminance levels. With the white sub-pixel always being active, as the TV becomes brighter the colour volume gets lower. This colour volume desaturation begins at around 45% digital stimulus and plateaus.

When you perform colour calibration in alternate software solutions, all you are doing is measuring the peak primary colours of your specific panel, and from that a new colour correction matrix is created to replace the factory one (hence, Matrix LUT). The 33x33x33 LUT slot is just used as a container.

The factory colour correction matrix is not the same as the factory calibration performed by LG Displays. It is a generic matrix that maps the generic native panel capabilities of WRGB OLED to the Rec2020 colour space.

In short, there is absolutely nothing wrong with this. LG know the capability of the displays they are making, and provided you use one of the "accurate" picture modes you can't realistically make it any better. You may get some improved measurement results, but you will cause image issues in doing so, most commonly colour banding.

If white balance is correct and the panel tracks 2.2 gamma when bypassing the conversion to ST.2084 PQ EOTF, then it is as good as you can get it in HDR. Just set it and forget it, don't let the marketing speak of others tell you otherwise!!!

Q: But hang on, you can calibrate Dolby Vision colours, what's different?

A: Dolby Vision uses a completely different system and has its own internal processing algorithm within the TV. It is not controlled by LG's internal processing. Dolby Vision requires completely different adjustments to both greyscale and colour mapping compared to HDR.

When you adjust greyscale/gamma tracking in Dolby Vision "relative mode", you must then upload a configuration file that contains the peak luminance (and black) and the coordinates of the primary colours. When you upload this configuration, the Dolby Processing engine then knows you have made adjustments.

With Dolby Vision being a closed proprietary system, it can get awfully complicated trying to explain most of what happens in detail. Rest assured, following the Dolby Vision guide provided within this pack is the correct procedure to follow, and it works very well.

Q: Is there anything else I should know?

A: Sure. I could probably sit here and write a 200 page document on the quirks of calibrating LG OLED with a 3D LUT. But that is not the purpose of these guides. The intention has only ever been to give new users a bit of a head start on what can be a somewhat overwhelming process when first beginning.

The rest is up to you. The information is out there, so if you want to know more then you must do the research.

It must be noted that I have no affiliation to Light Illusion. I have not been asked to provide these guides and I have not received anything for doing so. Everything has been done of my own free will in my own free time. It is my way of "paying back the community that gave to me" when I first began my journey into calibration. Ted Aspiotis was a tremendous resource of help and advice to me, as he has many before and since.

I have provided these guides to Ted to distribute to paid license holders of ColourSpace (HTL or higher) only, and I only ask that they are not redistributed in any way.

Useful Links:

- [ColourSpace CMS – Main AVS Forum Thread](#)
- [LightSpace CMS – LG OLED 3D LUT – AVS Forum Thread](#)
- [Light Illusion – ColourSpace Guides Index](#)
- [Light Illusion – i1 Display Pro Integration Guide](#)
- [Light Illusion – Video Guides](#)
- [Raspberry Pi PGenerator – Dedicated AVS Forum Thread](#)
- [Display Calibrations – Ted Aspiotis website](#)